

Tony Koo

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EDUCATION

NATIONAL UNIVERSITY OF SINGAPORE (NUS)

Aug 2023 - Jun 2026

Bachelor of Science in Business Analytics (Computing)

- Awards: University Engineering Scholarship, Sustainability Exchange Champion
- Projects: Smoodee, IBKR Algo Bot, P2P Loan Default Prediction System, Hull Tactical, NichGPT
- Relevant Coursework: Programming Methodology, Data Structures and Algorithms, Machine Learning, Quantitative Finance, Big Data & ML, Econometric Modelling, Information Systems Communications, BlockChain Technology

TEMASEK POLYTECHNIC

Apr 2018 - May 2021

Diploma in Accountancy & Finance

- Awards: Letters of Appointment: A&F Vice-President and Business EXCO, ISCA award
- Projects: Value at Risk, Securities & Asset Portfolio Management, WireCard, YouCard, Portz
- Relevant Coursework: Accounting, Economics, Financial and Risk Analysis, FinTech, Portfolio Management

WORK EXPERIENCE

TRADITION ENERGY *Quantitative Developer (Singapore, Singapore)*

May 2024 - Jul 2025

- Engineered automated yield curve construction workflows utilising interpolation techniques to maintain high-precision pricing for real-time complex financial derivatives.
- Synthesised complex methodologies from academic literature to architect and construct bespoke quantitative models proposed by clients, bridging the gap between theoretical research and production-ready solutions.
- Developed high-performance correlation analytical solutions for APAC markets leveraging Bloomberg and LSEG API integration; optimised calculation performance from 6.09s to 0.86s via advanced sorting algorithms.

TRADITION ENERGY SINGAPORE *Middle Office Analyst (Singapore, Singapore)*

Sep 2020 - May 2021

- Spearheaded onboarding of 20+ international clients, streamlining existing SOPs. Coordinated with trading desks and operations teams to ensure accurate execution of high-volume trades, streamlining process to reduce errors.
- Partnered closely with the London HQ and APAC offices to maintain smooth transactions across time zones, improving coordination and communication between global offices, reducing time-lag in work output.

SKEZI *Data Scientist (Paris, France)*

Jul 2025 - Dec 2025

- Architected advanced RAG pipelines with Transformers, utilising LangChain and FAISS-backed semantic caching to enhance outputs while cutting down LLM API costs and optimising latency via mean-pooled embeddings.
- Constructed a Computer Vision-based pipeline for automated PDF data extraction for medical questionnaire image extraction and processing with ~90% accuracy.
- Designed and implemented a proprietary Reinforcement Learning (RL) framework, the Self-RL loop, optimised for data efficiency by requiring a fraction of training data used in standard RL frameworks with 100% effectiveness.

PROJECTS

Healthcare Capstone Project, AI Engineer

Jan 2026 - Apr 2026

Agentic Collaborative Reasoning for the oncology domain

- Architected a LangGraph multi-agent orchestration system enabling clinicians to query healthcare data via NLP.
- Engineered a Feature Engineering Agent implementing the Refactor → Break Down → Compile decision procedure (ArXiv 2506.09742), improving Kaplan-Meier survival analysis reliability by eliminating column-name guessing.
- Designed a Human-in-the-Loop (HITL) workflow with clinician review checkpoints leveraging LangGraph interrupts, ensuring auditability before any results are delivered.

SKILLS, ACTIVITIES & INTERESTS

Technologies: Python, SQL, Java, R, VBA, API, Git, Tableau, Pandas, HuggingFace, LangChain/Graph, Scikit-learn, SciPy

AI & Engineering: RAG, Reinforcement Learning, Transformers, , PyTorch, TensorFlow, Prompt Engineering

Finance: HFT, Yield Curves, Monotone Convex, Risk Management, Derivatives Pricing, Portfolio Management, Derivatives

Certifications & Training: Chartered Financial Analyst (CFA) Foundations

Spoken Languages: English (Native), Chinese Mandarin (Intermediate), French (Basic)